

AWS Certified Developer - Associate

Study Guide

Short-Answer Quiz

1. Explain the concept of "Envelope Encryption" within AWS Key Management Service (KMS) and describe the roles of the Customer Master Key (CMK) and the Data Key.
 2. What are the key differences between AWS Systems Manager Parameter Store and AWS Secrets Manager in terms of their primary use cases and features?
 3. Describe the two primary methods for querying data in DynamoDB, "Query" and "Scan," and explain which operation is generally more efficient and why.
 4. What is the purpose of an Elastic Load Balancer (ELB), and what are the distinct operational layers and typical use cases for an Application Load Balancer versus a Network Load Balancer?
 5. Define the function of a Dead-Letter Queue (DLQ) in the context of AWS Lambda and explain how it contributes to error handling for asynchronous invocations.
 6. What is an `AppSpec.yml` file in AWS CodeDeploy? Describe its purpose and list at least two types of information it defines for a deployment.
 7. Compare and contrast the "Eventually Consistent Reads" and "Strongly Consistent Reads" models in Amazon DynamoDB, highlighting the trade-offs between them.
 8. Explain the difference between S3 Standard-Infrequent Access (S3-IA) and S3 One Zone-Infrequent Access storage classes, focusing on durability and cost.
 9. What is the primary function of AWS CloudFormation, and what is the relationship between a CloudFormation template and a stack?
 10. Describe the difference between the "In-Place" and "Blue/Green" deployment approaches available in AWS CodeDeploy.
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Quiz Answer Key

1. Envelope encryption is the practice of encrypting plaintext data with a data key and then encrypting that data key under another key for protection. The Customer Master Key (CMK), now called a KMS Key, is used to generate, encrypt, and decrypt the Data Key. The Data Key is then used to encrypt and decrypt the actual application data, which can be larger than the 4KB limit for CMK operations.
2. Parameter Store is designed to store configuration data and less-sensitive secrets like API keys, supporting plaintext and encrypted values without automatic rotation. Secrets Manager is built specifically for highly sensitive credentials like database passwords,

encrypting all secrets by default and providing features for automatic rotation, auditing, and fine-grained access control.

3. A "Query" operation in DynamoDB finds items based on a primary key value and is highly efficient because it goes directly to the data. A "Scan" operation examines every item in the entire table, making it less efficient and potentially slower. A Query should always be used if possible to improve performance.
4. An Elastic Load Balancer (ELB) distributes network traffic across a group of servers. An Application Load Balancer operates at Layer 7 (HTTP/HTTPS) and is used for web traffic, while a Network Load Balancer operates at Layer 4 (TCP) for high-performance workloads that are typically not web applications.
5. A Dead-Letter Queue (DLQ) is an error-handling feature for AWS Lambda that captures and saves failed asynchronous invocation events for further processing or debugging. After Lambda automatically retries a failed invocation twice, it sends the event to a specified SQS queue or SNS topic. This ensures that failed events are not lost and can be analyzed.
6. The `AppSpec.yml` file is a configuration file used by CodeDeploy that defines the parameters for a deployment. It specifies the location of application files to be deployed (the `files` section) and defines scripts that need to be run at specific points in the deployment lifecycle (the `hooks` section), such as `BeforeInstall` or `ApplicationStart`.
7. Eventually Consistent Reads, the default model, means that all copies of data will be consistent within about a second, offering the best read performance. Strongly Consistent Reads reflect all successful writes instantly, ensuring the most up-to-date data is read but at the cost of potentially higher latency.
8. S3 Standard-IA is for data accessed less frequently but requiring rapid access, stored across at least three Availability Zones. S3 One Zone-IA is similar but stores data redundantly within a single Availability Zone, making it 20% cheaper but less durable, suitable for non-critical, reproducible data.
9. AWS CloudFormation is a service that allows you to manage, configure, and provision AWS infrastructure as code using YAML or JSON files. A CloudFormation template is the file where you define your AWS resources, and a stack is the set of resources that CloudFormation creates based on that template.
10. An "In-Place" deployment stops the application on each instance, installs the new version, and restarts it, which causes a reduction in capacity during the update. A "Blue/Green" deployment provisions a new set of instances, installs the new release on them, and then shifts traffic over from the old "blue" environment to the new "green" environment, avoiding downtime.

Essay Questions

1. An e-commerce application is being developed with a serverless architecture. Describe a potential workflow using API Gateway, Lambda, DynamoDB, SQS, and SNS for a customer placing an order. Explain the role each service plays and how they interact to create a loosely coupled, event-driven system.
 2. Discuss the complete CI/CD workflow on AWS. Detail the specific roles of CodeCommit, CodeBuild, CodeDeploy, and CodePipeline in moving a code change from a developer's local machine to a production environment.
 3. Compare the different data storage options available for a cloud-native application, including S3, EBS, RDS, and DynamoDB. Analyze the primary characteristics of each (e.g., object vs. block, SQL vs. NoSQL) and provide a specific use case where each would be the optimal choice.
 4. A developer needs to secure a web application running on EC2 instances behind an Elastic Load Balancer. Detail a multi-layered security strategy that incorporates IAM roles, S3 bucket policies for data storage, KMS for encryption, and Cognito for user authentication.
 5. Explain the monitoring and troubleshooting capabilities provided by the combination of Amazon CloudWatch, AWS CloudTrail, and AWS X-Ray. Describe what kind of information each service provides and how a developer would use them together to perform a root cause analysis on a distributed application.
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Glossary of Key Terms

Term	Definition
Access Control Lists (ACLs)	A security feature for S3 that can be applied at the object level to define which AWS accounts or groups are granted access to specific objects.
Alias Record (Route 53)	A type of Route 53 record that allows you to route traffic to an AWS resource (like an ELB or S3 bucket) at the top of a DNS namespace (the zone apex), functioning like an A Record but without using an IP address.
Amplify	A set of tools and services designed to make it easy to develop web and mobile applications on AWS, handling backend development and providing fully managed web hosting with CI/CD.

API Gateway	A managed service that allows developers to publish, maintain, monitor, and secure APIs. It supports RESTful and WebSocket APIs and can connect to backends like Lambda, EC2, and DynamoDB.
AppSpec File	A configuration file, in YAML or JSON, used by AWS CodeDeploy to define deployment parameters, including file locations and lifecycle event hooks.
Application Load Balancer (ALB)	A type of Elastic Load Balancer that operates at Layer 7 (HTTP/HTTPS) to distribute network traffic. It is ideal for web applications.
Athena	An interactive query service that enables you to run standard SQL queries on data stored directly in Amazon S3 without needing to provision infrastructure.
Automated Backup (RDS)	A feature enabled by default in RDS that allows for point-in-time recovery of a database within a specified retention period. Backups are stored in S3.
Canary Testing	A deployment strategy where a small percentage of incoming traffic is directed to a new application version for a testing period before a full rollout. Elastic Beanstalk's "Traffic Splitting" deployment enables this.
Classic Load Balancer	A legacy Elastic Load Balancer that supports both HTTP/HTTPS and TCP traffic.
Cloud Development Kit (CDK)	An open-source software development framework for defining cloud infrastructure in code using familiar programming languages, which then synthesizes into CloudFormation templates.

CloudFormation	An AWS service for managing, configuring, and provisioning infrastructure as code using YAML or JSON templates. A set of resources created from a template is called a stack.
CloudFront	AWS's Content Delivery Network (CDN) that caches content at distributed Edge Locations to deliver webpages and other content to users with low latency.
CloudTrail	A service that records user activity and API calls within an AWS account, providing an audit log for creation, modification, or deletion of resources.
CloudWatch	A monitoring service for AWS resources and applications, collecting metrics, logs, and events to provide performance and health insights.
CodeArtifact	An artifact repository service for storing, publishing, and sharing software packages required for projects, integrating with both internal and public repositories.
CodeBuild	A fully managed continuous integration service that compiles source code, runs tests, and produces software packages that are ready to deploy.
CodeCommit	A fully managed source control service, based on Git, that hosts secure and private repositories for source code, binaries, libraries, and images.
CodeDeploy	A service that automates code deployments to instances like EC2 or services like Lambda, supporting deployment strategies such as in-place and blue/green.
CodePipeline	A fully managed continuous delivery service that automates the build, test, and deploy phases of a release process every time there is a code change.

Cognito	A service that provides user sign-up, sign-in, and access control for web and mobile applications. It acts as an identity broker for web identity federation.
CORS (Cross Origin Resource Sharing)	A mechanism that allows resources in one S3 bucket to access resources in another S3 bucket.
Customer Master Key (CMK)	Now called a KMS Key, this is a logical representation of a master key in KMS used to generate, encrypt, and decrypt data keys for envelope encryption.
Data Key	An encryption key generated by a CMK in KMS, used to encrypt and decrypt application data outside of KMS.
Dead-Letter Queue (DLQ)	An SQS queue or SNS topic designated to receive and store messages from another queue or Lambda function that have failed processing, allowing for later analysis.
DynamoDB	A fully managed, serverless, low-latency NoSQL database that supports key-value and document data models.
DynamoDB Accelerator (DAX)	A fully managed, in-memory cache for DynamoDB that delivers up to a 10x read performance improvement for read-heavy workloads.
DynamoDB Streams	A feature that captures a time-ordered sequence of item-level modifications in a DynamoDB table and stores this log for 24 hours, useful for auditing or triggering events.

EC2 (Elastic Compute Cloud)	A web service that provides secure, resizable compute capacity (virtual machines) in the cloud.
EBS (Elastic Block Store)	A high-performance block storage service designed for use with Amazon EC2 for both throughput and transaction-intensive workloads.
EBS Snapshot	A point-in-time copy of an EBS volume, used for backups or to create new EBS volumes.
Edge Locations	A network of over 200 worldwide locations where CloudFront caches copies of web content to reduce latency for end-users.
EFS (Elastic File System)	A scalable, elastic, cloud-native file storage service for use with AWS Cloud services and on-premises resources. It can be mounted by Lambda functions in the same VPC.
ElastiCache	An in-memory cache service that improves database performance for read-heavy workloads. It supports two engines: Memcached and Redis.
Elastic Beanstalk	A service for deploying and scaling web applications and services. It handles infrastructure management, including load balancing, auto-scaling, and health monitoring.
Envelope Encryption	The practice of encrypting plaintext data with a data key and then encrypting the data key itself with another key (like a CMK).
EventBridge	A serverless event bus that makes it easy to connect applications together using data from your own applications, SaaS applications, and AWS services.

Exponential Backoff	An algorithm that progressively increases wait times between consecutive retries of a failed request to avoid overwhelming a service. AWS SDKs often implement this automatically.
FIFO (First-In, First-Out) Queue	A type of SQS queue that preserves the exact order in which messages are sent and received and ensures a message is processed exactly once.
IAM (Identity and Access Management)	A global AWS service that helps you securely control access to AWS resources by managing users, groups, roles, and their permissions.
IAM Group	A collection of IAM users. Permissions applied to a group are inherited by all users within that group.
IAM Policy	A JSON document that formally states one or more permissions. It can be attached to a user, group, or role.
IAM Role	An IAM identity with permission policies that determine what the identity can and cannot do in AWS. It can be assumed by users, applications, or services.
IAM User	An end user (such as a person, application, or service) that interacts with AWS.
Kinesis	A family of services for collecting, processing, and analyzing real-time streaming data at a massive scale.

KMS (Key Management Service)	A managed service that makes it easy to create and control the encryption keys used to encrypt data.
Lambda	A serverless compute service that lets you run code without provisioning or managing servers. It executes code in response to events and automatically manages the compute resources.
Lambda Layers	A distribution mechanism for libraries, custom runtimes, and other function dependencies. They allow you to manage your in-development function code independently from the unchanged code and resources that it uses.
MemoryDB for Redis	A Redis-compatible, durable, in-memory database service that delivers ultra-fast performance. It can be used as a primary database.
Multi-AZ (RDS)	A high-availability feature for RDS that creates an exact copy of a database in another Availability Zone. AWS automatically fails over to this standby instance in case of an outage.
Network Load Balancer (NLB)	A type of Elastic Load Balancer that operates at Layer 4 (TCP) and is capable of handling millions of requests per second with high performance and low latencies.
OLAP (Online Analytics Processing)	A category of software tools that analyze data stored in a database. RDS is generally not suitable for OLAP workloads; a service like Redshift is preferred.
OLTP (Online Transaction Processing)	A category of data processing that is focused on transaction-oriented tasks. RDS is well-suited for OLTP workloads.

Origin (CloudFront)	The source of the files that a CloudFront distribution will serve. An origin can be an S3 bucket, an EC2 instance, an Elastic Load Balancer, or Route 53.
Parameter Store	A capability of AWS Systems Manager that provides secure, hierarchical storage for configuration data and secrets management.
Read Replica (RDS)	A read-only copy of a primary database. It can be used to offload read traffic and can be promoted to become an independent database if needed.
Route 53	Amazon's highly available and scalable cloud Domain Name System (DNS) web service.
S3 (Simple Storage Service)	A secure, durable, and highly scalable object storage service. It stores data as objects within resources called buckets.
S3 Bucket Policy	A resource-based policy written in JSON that specifies what actions are allowed or denied for an entire S3 bucket.
S3 Glacier	A family of low-cost S3 storage classes (Instant Retrieval, Flexible Retrieval, Deep Archive) intended for data archiving and long-term backup.
S3 Intelligent-Tiering	An S3 storage class that automatically moves objects between a frequent access tier and an infrequent access tier to optimize storage costs based on access patterns.
Secrets Manager	A secrets management service that helps protect access to applications, services, and IT resources by enabling rotation, management, and retrieval of credentials.

Serverless Application Model (SAM)	An open-source framework and extension for CloudFormation that provides a simplified way of defining and deploying serverless applications.
SNS (Simple Notification Service)	A fully managed pub/sub messaging service that enables you to decouple microservices, distributed systems, and serverless applications.
SQS (Simple Queue Service)	A fully managed message queuing service that enables you to decouple and scale microservices, distributed systems, and serverless applications. It is a pull-based service.
Step Functions	A serverless function orchestrator that makes it easy to sequence AWS Lambda functions and multiple AWS services into business-critical applications through a visual interface.
Web Identity Federation	A feature that allows users to access AWS resources after authenticating with a web-based identity provider, such as Amazon, Facebook, or Google, often brokered by Amazon Cognito.
X-Ray	An AWS service that helps developers analyze and debug distributed applications, such as those built using a microservices architecture, by providing a visualization of the application's components.